# Classification

## Decision Tree

Advantages:

- Handling missing data
- Easy to implement and interpret
- They are fast

Disadvantage:

- Greedy selection is not **accurate** → usually end up going to maximum depth

Cost:

- $O(ndk)$ for stump (n data, d feature, k threshold)
  - Binary Features: $O(ndk)$
  - Numerical Features: $O(n^2d)$or if improved $O(ndlog(n))$

## Probabilistic Classifiers

- Naïve Bayes:
  - $P(y \mid x_1, \ldots, x_n) = \dfrac{P(y)\prod_{i=1}^{n} P(x_i \mid y)}{P(x_1, \ldots, x_n)}$
  - $P(x_1, \ldots, x_n \mid y) = \prod_{i=1}^{n} P(x_i \mid y)$
  - Laplace Smoothing: $P(x_i = c \mid y) = \dfrac{\text{Count}(x_i = c \text{ and } y) + \beta}{\text{Count}(y) + \beta \cdot k}$

    where $x_i$has $k$ levels.
- Decision Theory: Used when false positive/negative costs differently; giving different costs to false positive and false negative.
- Cost:
  - $O(nd)$

## Regression

$\hat{y} = \text{sign}(o_i)$

- Objectives
  1. $\sum ||\text{sign}(o_i) - y||_0$
     - $o_i = w^T x_i$
     - it is non-convex and hard to optimize
  2. 0-1 loss:
     - $\sum \max(0, -o_i y_i)$
  3. Hinge Loss (not degenerative version of 0-1 loss)
     - $\sum \max(0, 1 - o_i y_i)$
  4. Support Vector Machine:
     - $\sum \max(0, 1 - o_i y_i) + \frac{\lambda}{2}||w||^2$
  5. Logistic Loss:
     - $\sum \log(\exp(0) + \exp(-o_i y_i)) = \sum \log(1 + \exp(-o_i y_i))$

$\hat{y} = p(y = c|x) = \dfrac{1}{1 + \exp(-o_i)}$

## Non-parametric Models → model grows with data

1. **K-nearest Neighborhood (KNN)**
   - Scaling Features is important
   - Distance Measures:
     - $L_1$ or "Manhattan" norm. $|r|_1 = |r_1| + |r_2|$
     - $L_2$ or "Euclidean" norm. $|r|_2 = \sqrt{r_1^2 + r_2^2}$
     - $L_\infty$ or "max" norm. $|r|_\infty = \max |r_1|, |r_2|$
   - Cost:
     - $O(nd)$

## Hierarchical Clustering - Agglomerative

- Distance between means
- Cost
  - $O(n^3d)$
- Dendrogram

# Outlier Detection

### Outlier Detection

1. Model-based: z-score
   - Disadvantages:
     - z-score is sensitive to outlier
     - It assumes the data is uni-modal
2. Graphical
   - Disadvantages:
     - Only two variables at a time
3. Cluster-based:
   - K-mean clustering
     - Clusters with few points
   - Density-based clustering:
     - Points with not assignments
     - Disadvantages:
       - different densities
4. Distance-based:
   - KNN → average of distance to KNN
   - Disadvantages:
     - different densities →
       - solution: outlierness - ratio between the average of distance to the neighbors/their average to their neighbors
5. Supervised-learning:
   - Disadvantages:
     - Needs a model to make the labels first

# Unsupervised - Parametric

### Parametric Models

1. **K-mean Clustering**
   - Advantages
     - K-means is guaranteed to converge for Euclidean distance
   - Disadvantages
     - It may converge to sub-optimal solution (local optimum) → sensitive to initialization
     - Cluster regions are always convex → cannot model non-convex clusters

Cost:

- Assigning Clusters ~ $O(ndk)$
- Updating Means ~ $O(nd)$

# Unsupervised – Non-Parametric

### Density-Based Clustering

- $\epsilon$: distance of a neighbor
- Min Neighborhoods: Number of neighborhood for a dense region
- Advantages
  - Can cluster non-convex
- Disadvantages
  - Non-assigned points
  - Sensitive to be selection for parameters
  - Finding cluster is expensive → non-parametric model
  - Clusters with different densities
- Cost:
  - $O(n^2d)$

## Linear Algebra

$$||r||^2 = r^T r$$
$$(AB)^T = B^T A^T$$
$$a^T Ab = b^T A^T a$$

$$\nabla[c] = 0 \text{ (zero vector)}$$
$$\nabla[w^T b] = b$$
$$\nabla[\frac{1}{2}w^T Aw] = Aw \text{ (if A is symmetric)}$$
$$\nabla[f(g(x))] = (\nabla g(x))^T(\nabla f(g(x)))$$

## Linear Regression

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{id} \end{bmatrix}$$

$\hat{y}_i = w^T x_i$ for one single data point (vector form)

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix} = \begin{bmatrix} — x_1^T — \\ — x_2^T — \\ \vdots \\ — x_n^T — \end{bmatrix}$$

$\hat{y} = Xw$ for all data points (matrix form)

## Non-linear Regression - Basis

### Matrix Notation:

- Replace $X$ with $Z$ (for training and testing)

$$Z = \begin{bmatrix} 1 & x_1 & (x_1)^2 & \cdots & (x_1)^p \\ 1 & x_2 & (x_2)^2 & \cdots & (x_2)^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & (x_n)^2 & \cdots & (x_n)^p \end{bmatrix}$$

## Non-linear Regression – non-parametric

Radial Basis Function (Gaussian RBF)

- $g(\epsilon) = \exp(-\epsilon^2/(2\sigma^2))$
- $\sigma$ is a hyper-parameter
- Replace $X(n \times d)$ with $Z(n \times n) \rightarrow$ add bias by adding column on 1

$$Z = \begin{bmatrix} g(||x_1 - x_1||) & g(||x_1 - x_2||) & \cdots & g(||x_1 - x_n||) \\ g(||x_2 - x_1||) & g(||x_2 - x_2||) & \cdots & g(||x_2 - x_n||) \\ \vdots & \vdots & \ddots & \vdots \\ g(||x_n - x_1||) & g(||x_n - x_2||) & \cdots & g(||x_n - x_n||) \end{bmatrix}$$

- Replace testing data $\hat{X}(t \times d)$ with $\hat{Z}(t \times n)$
- Cost
  - O(nd) for keeping data

## Gradient Descent

### Formula

$$w^{t+1} = w^t - \alpha^t \nabla f(w^t)$$

### Cost

- $O(ndt) \rightarrow t$ is the number of iterations

## Feature Selection

### Forward Selection

- Score (similar to AIC)
  - $f(w) = \frac{1}{2}||Xw - y||^2 + \lambda||w||_0$
- Cost:
  - All Combinations $O(2^d) \times O(\text{each model})$
  - Greedy $O(d^2) \times O(\text{each model})$
  - Greedy for Least Square $\rightarrow O(nd^4 + d^5) \rightarrow$ Too Expensive

### L1-Regularization

- Advantages:
  - **Very fast alternative for Forward Selection**
  - L1-regularization give sparsity (sparsity is when coefficients are **zero**)
- Disadvantages:
  - No closed-form solution

## Convex Function

1. Area above the function be convex
2. Second derivative always $\geq 0$
3. Norm and Squared Norms are convex $\rightarrow$ except zero norm which is not convex
4. Sum of convex functions is a convex function
5. Max of convex functions is convex
6. Composition of convex function and a linear function is convex
   - $g(Xw - y)$ is convex if $g(x)$ is convex
   - composition of a convex function with another convex function is not convex
7. Multiplication of convex functions are NOT convex

## Least Square

$$LS := f(w) = \frac{1}{2}||Xw - y||^2 = \frac{1}{2}w^T\underbrace{(X^T X)}_{A}w - w^T\overbrace{(X^T y)}^{b} + \frac{1}{2}y^T y$$
$$\rightarrow \nabla f(w) = X^T Xw - X^T y = 0$$

Cost:

- Forming matrices: $O(n^2 d)$
- Solving matrices: $O(d^3)$
- Total: $O(nd^2 + d^3)$

Disadvantages:

- It is computational expensive if $d$ is large $\rightarrow$ Gradient Descent
- Solution may not be unique
- Sensitive to outlier
- Data might be big to store $(X^T X)$

## Other

$$\sum_{i=1}^{n} c_i x_i = c^T x (\text{or } x^T c)$$
$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij} x_i x_j = x^T A x$$
$$\sum_{i=1}^{n} x_i r_i = x^T r$$

**Norm** $||$

$$\sum_{i=1}^{n} |r_i| = ||r||_1$$
$$\sum_{i=1}^{n} r_i^2 = ||r||^2 \rightarrow ||r||_2 = \sqrt{\sum_{i=1}^{n} r_i^2}$$
$$\max_{i \in \{1,2,\ldots,n\}} \{|r_i|\} = ||r||_\infty$$